



McQueen's Lifesaver

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Problem Statement & Inspiration

What's the Problem?

- Faulty tires could cause serious safety issues to a racecar driver.
- Since 2022, every F1 race car has carried a standard FIA tyre-pressure sensor - no analytics.
- Public F1 data do not directly show every tire's pressure, temperature, or wheel speed.

What did we do?

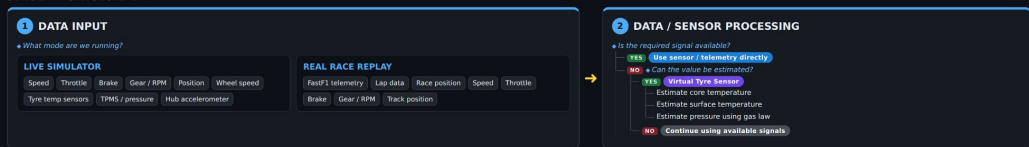
- Lightning Response watches every tyre in real time.
- Detects: lock-ups, wheelspin, overheating, cold tyres, flat spots and air loss
- Predicts how many laps each tyre has left, and radios the pit crew's phones before accidents.
- Turns available telemetry and historical stint data into earlier, per-tyre risk warnings
- Easier to address the root cause before a safety hazard.

⚡ LIGHTNING RESPONSE

Tyre-risk decision pipeline — from raw telemetry to pit-wall call, human response, and verification

Decision / question YES NO branch OK / normal watch / manage box this lap critical process / output

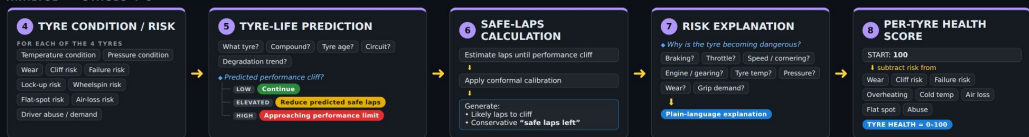
SENSE · STAGES 1-2



DETECT · STAGE 3 (RUNS IN PARALLEL)



ANALYSE · STAGES 4-8



DECIDE · STAGES 9-11



ACT & LEARN · STAGES 12-14



Ingest & Clean

**Detect Risk triggers and
Predict**

Assess and Act

Feedback Loop

ollon

AMPERE

TELUS

**W
O
R
K
F
L
O
W**

Tech Stack

Backend

- Python 3.11+, FastAPI, Uvicorn
- WebSockets connect the pit wall, driver phone and crew phone
- In-memory state only (no database, queue or auth)
- It generates the QR codes that open the pages on phones`n

Frontend

- Plain HTML, CSS and JavaScript, with no framework and no build step
- Canvas 2D and SVG draw the track map, car and charts
- Vibration API alerts the crew phone
- Google Fonts: Inter and JetBrains Mono

Tooling

- Pytest for tests
- Uv / pip for Python dependencies (pyproject.toml, uv.lock)
- Git and GitHub

Machine learning and data

- Pandas, NumPy, SciPy (signal smoothing) and PyArrow (Parquet files)
- LightGBM with monotone constraints and quantile models, for tyre life and lock-up/wheelspin risk
- Scikit-learn for logistic regression, calibration and race-grouped cross-validation
- Joblib stores the trained models (models/weights/)
- Models: gas law for the virtual tyre-pressure sensor, a CUSUM leak detector, and a tyre and car simulator

Data sources (6 in total)

- FastF1 / OpenF1: real 2019–2025 lap (skipped COVID years), tyre, weather and telemetry data
- Assetto Corsa Gym (from Hugging Face): simulator data with per-wheel slip, used as ground truth
- F1 tyre-strategy dataset from Kaggle

Lightning Response: impact in numbers

AI tyre-safety pit wall · FormulaTech Hacks

Warns before it happens

tested on drivers the models never saw

0.5 s

median warning before a lock-up

1.75 s

median warning before wheelspin

2-7 x

a driver's 0.25 s reaction time

85%

of lock-ups caught early

99%

of wheelspins caught early

0.96 / 0.98

ROC-AUC, lock-up / wheelspin

1 in 5

really happens when risk reads 20%

Proven on real F1 races

a season the models never saw

15 laps

BOX before Hamilton's 2020 tyre failure

0.87

tyre-cliff prediction score, 2025

93%

of "safe laps left" promises held

2 x

fewer broken promises than uncalibrated

33

circuit stint limits, set from data

14-30

laps per set (FIA set 18 at Qatar)

17%

lower lap-time error than a constant

Sees what public data can't

virtual sensors and physics

6-9 °C

tyre-temperature error, unseen track

40-50%

lower error than a naive estimate

0.98 psi

tyre-pressure error, from the gas law

~1.6 s

to flag air loss after a puncture

99%

of cold-tyre moments caught, 3 s early

x11

lock-up odds on overheated tyres

Live, and at scale

measured on the current build

4 / s

risk updates, physics in real time

0.3-0.4 s

for a BRACE alert to reach the driver

4 s

early lock-up warning, live demo

11 → 0

wheelspin alerts in tidy driving, per 150 s

138,683

real F1 laps from 127 races

767,253

simulator moments with wheel data

287

automated tests passing

Model results are on data the models never trained on (unseen drivers, track or season). Early warnings: Assetto Corsa Gym, 116 stints. Live: Lightning Response simulator.

Models

Tier A: Can we detect an event before it happens?

- Classification Strength: ROC-AUC / PR-AUC
- Operational Warning Performance: detection rate, lead time, safe-lap violation rate

Risk Model: Does the risk happen? What factors caused it?

- Post-hoc Calibration: binned predictions, predicted probability vs observed probability

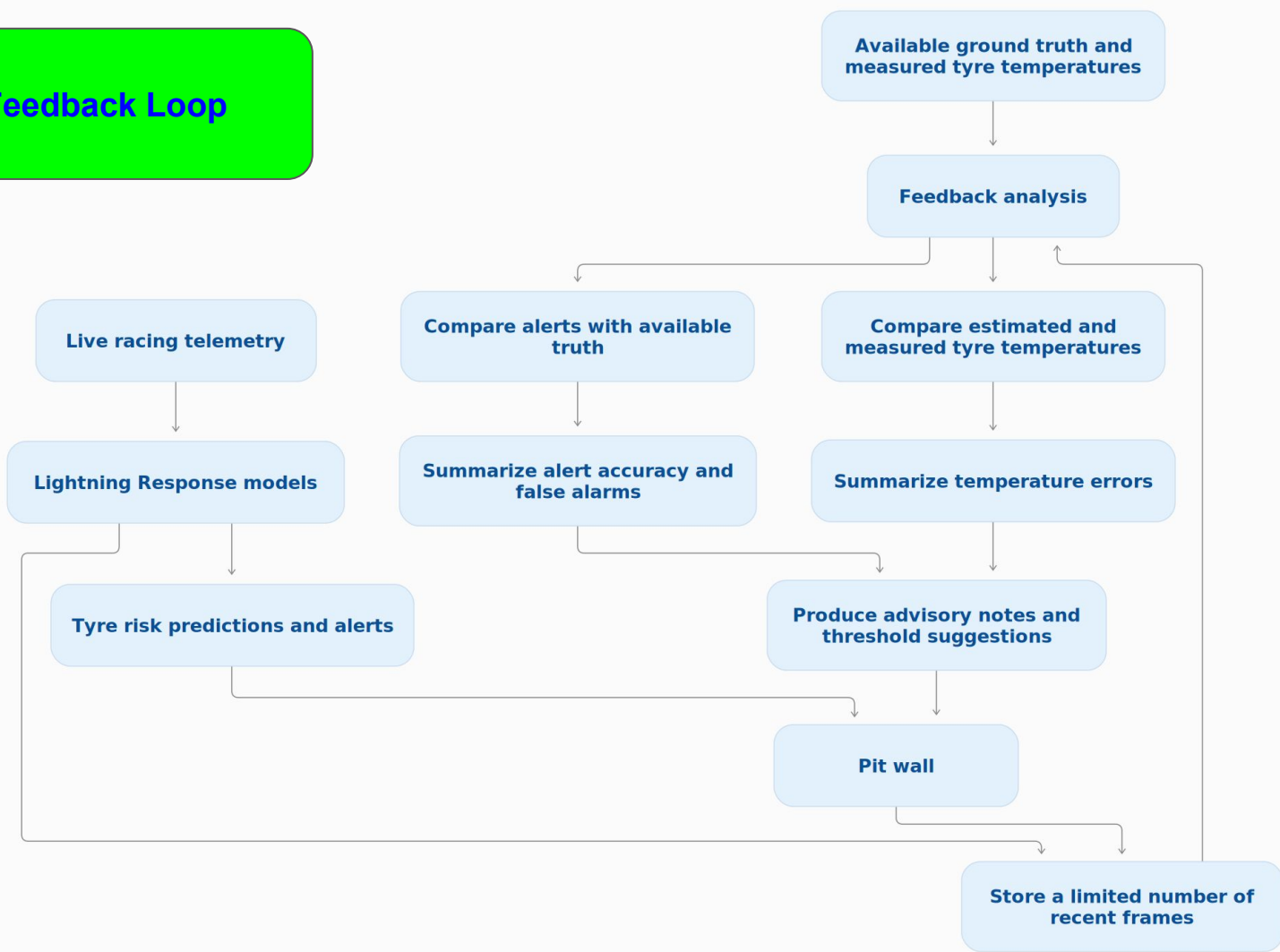
Tier B: Can we predict tyre degradation/failure farther into the stint? How conservative is our safe-lap estimate?

- ROC-AUC / PR-AUC, event detection rates, held-out generalization gap, conformal prediction, violation rates, predicted safe laps, binned predictions, alert thresholds.

Virtual Telemetry: How close are our estimated tyre temperatures and pressure to the measured values?

- Regression Error: temperature and tire pressure MAE, benchmarked

The Feedback Loop





⚡ LIGHTNING RESPONSE

From raw data to the right pit call — in five steps

